NGS Library QC

General Information

Order Number	HN00280302	Name of Customer	EULALIO SANTOS	Date of Order	2026-06-26

Final QC Result of DNA sample(s)					
Arrival Date	Experiment Date	Sample count	Pass	Fail	Hold
N/A	N/A	N/A	N/A	N/A	N/A

Final QC Result of RNA sample(s)					
Arrival Date	Experiment Date	Sample count	Pass	Fail	Hold
2026-06-26	2026-07-14	14	12	2	0

*** Pass : Proceed with the library construction.**
*** Fail : Further processes are on hold until the replacement samples received.**
We do not recommend in proceeding further steps until a specific instruction was given from the client.
*** Hold : A specific instruction should be given by the client for further processing as the QC pattern may be triggered by the sample nature.**
Macrogen does not proceed the next step until we have received your permission.
As 3 ul was taken from the sample for sample (library) QC purposes, the indicated volume represents 3ul less than the total volume received.

Library QC Result of RNA

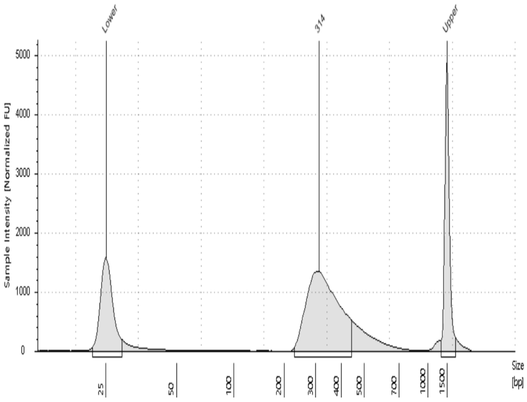
Arrival Date	2026-06-26	Experiment Date	2026-07-14	Tested by	CNY
Comment					

#	Library Name	Library Type	Conc. (ng/ul)	Conc. (nM)	Size (bp)	Result*	
1	ID-1	Illumina Stranded mRNA_HQ	41.28	202.26	314	Pass	
2	ID-2	Illumina Stranded mRNA_HQ	6.95	34.14	313	Pass	
3	ID-3	Illumina Stranded mRNA_HQ	7.45	35.27	325	Pass	
4	ID-5	Illumina Stranded mRNA_HQ	13.79	66.93	317	Pass	
5	ID-7	Illumina Stranded mRNA_HQ	2.16	10.73	309	Pass	
6	ID-8	Illumina Stranded mRNA_HQ	1.6	7.44	331	Fail	Low Quantity to (Run or Capture)
7	ID-9	Illumina Stranded mRNA_HQ	5.77	26.99	329	Pass	
8	ID-10	Illumina Stranded mRNA_HQ	5.19	27.15	294	Pass	
9	ID-12	Illumina Stranded mRNA_HQ	14.16	69.59	313	Pass	
10	ID-14	Illumina Stranded mRNA_HQ	12.16	61.34	305	Pass	
11	ID-15	Illumina Stranded mRNA_HQ	8.93	43.74	314	Pass	
12	ID-16	Illumina Stranded mRNA_HQ	6.3	29.09	333	Pass	

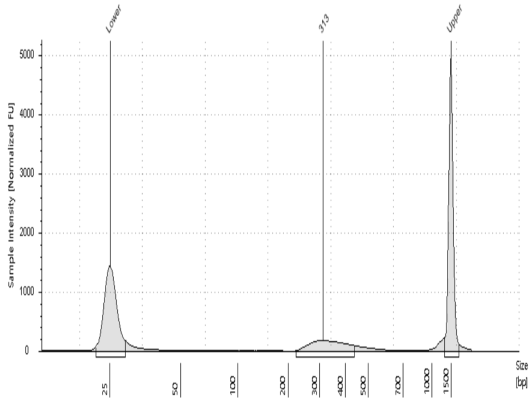
#	Library Name	Library Type	Conc. (ng/ul)	Conc. (nM)	Size (bp)	Result*	
13	ID-18	Illumina Stranded mRNA_HQ	6.26	29.46	327	Pass	
14	ID-21	Illumina Stranded mRNA_HQ	1.81	8.59	325	Fail	Low Quantity to (Run or Capture)

Experiment Condition	TapeStation D1000 Screen Tape

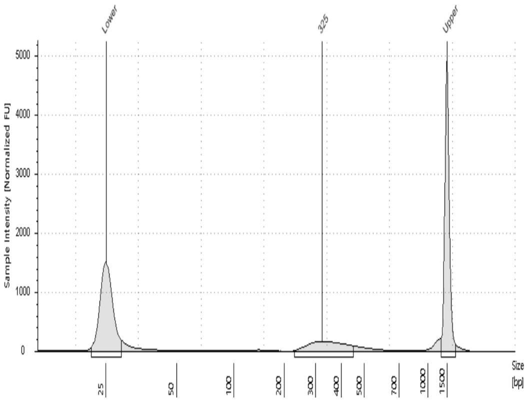
Click to Enlarge =>1:Library : ID-1



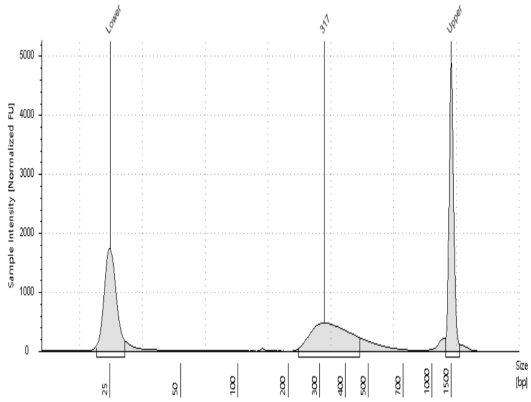
Click to Enlarge =>2:Library : ID-2



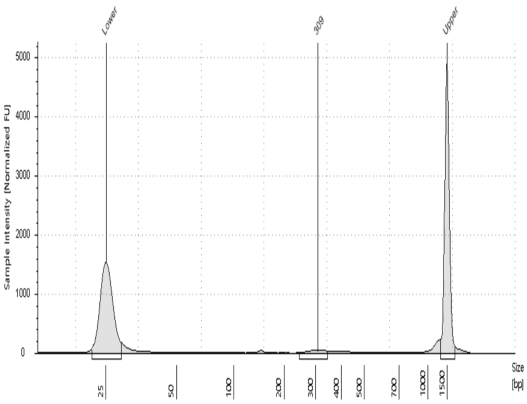
Click to Enlarge =>3:Library : ID-3



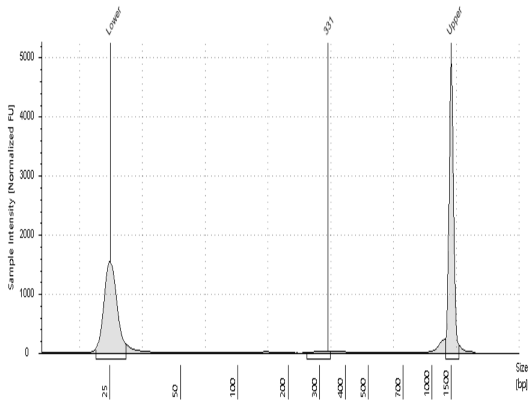
Click to Enlarge =>4:Library : ID-5



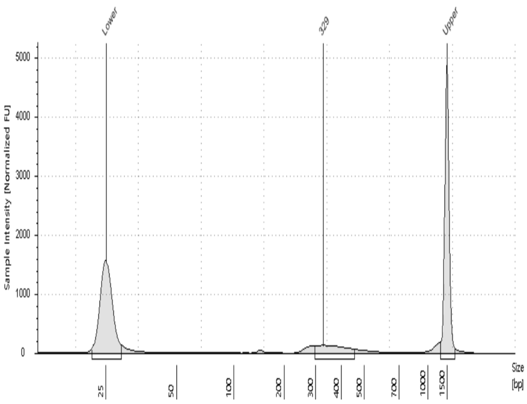
Click to Enlarge =>5:Library : ID-7



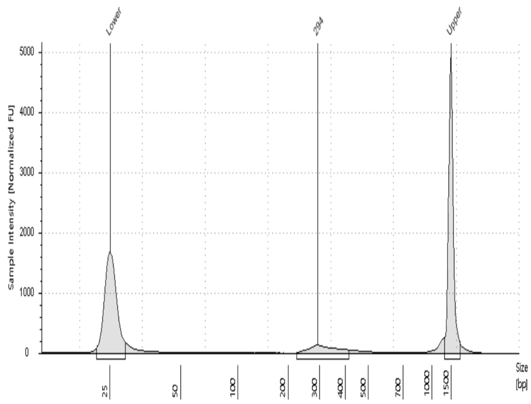
Click to Enlarge =>6:Library : ID-8



Click to Enlarge =>7:Library : ID-9

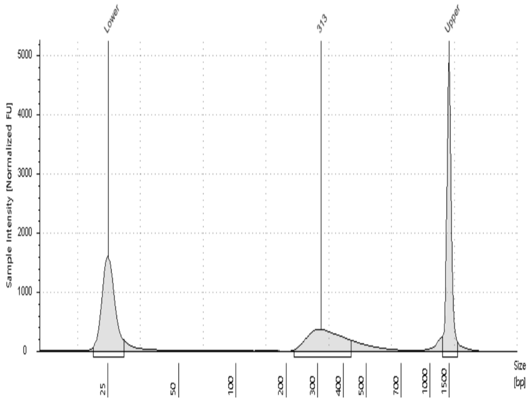


Click to Enlarge =>8:Library : ID-10

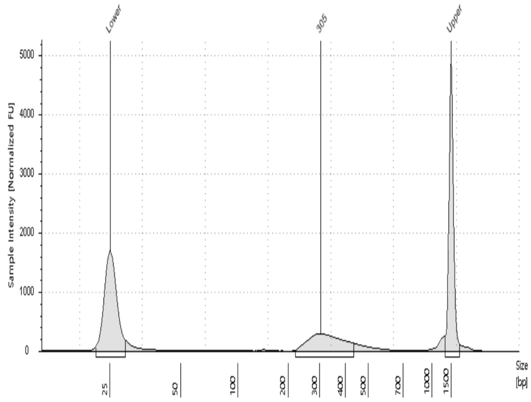


Experiment Condition	TapeStation D1000 Screen Tape

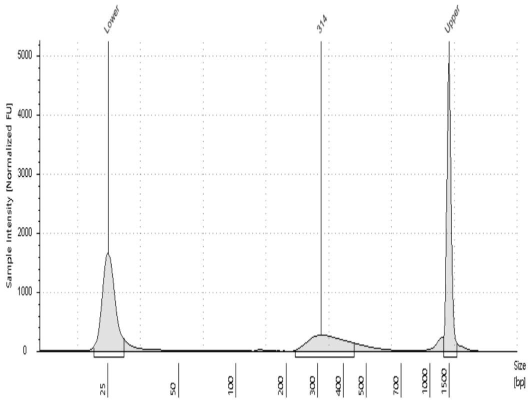
Click to Enlarge =>9:Library : ID-12



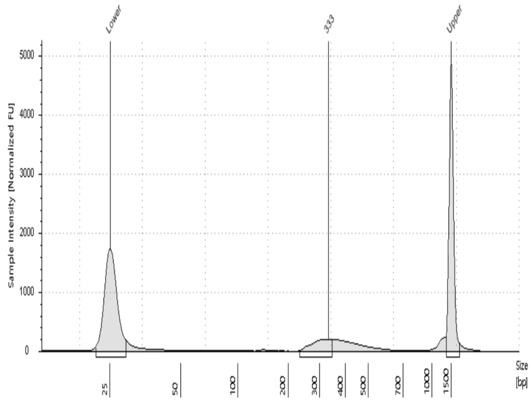
Click to Enlarge =>10:Library : ID-14



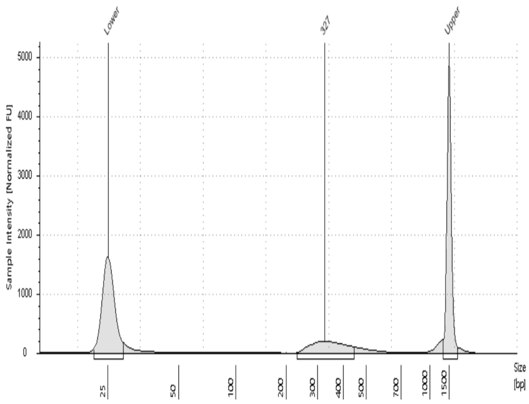
Click to Enlarge =>11:Library : ID-15



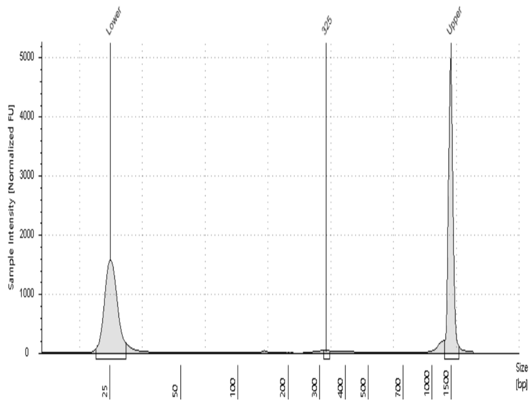
Click to Enlarge =>12:Library : ID-16



Click to Enlarge =>13:Library : ID-18



Click to Enlarge =>14:Library : ID-21



Library QC Method

1. Library Size Check

To verify the size of PCR enriched fragments, we check the template size distribution by running on a Agilent Technologies 2100 Bioanalyzer using a DNA 1000 chip.

2. Library Quantity Check

1) illumina library

In order to achieve the highest quality of data on Illumina sequencing platforms, it is important to create optimum cluster densities across every lane of every flow cell.
This requires accurate quantitation of DNA library templates.
So we quantify prepared libraries using qPCR according to the Illumina qPCR Quantification Protocol Guide.

2) PacBio library

To generate a standard curve of fluorescence readings and calculate the library sample concentration, we use Qubit standard Quantification solution and calculator.

Library RNA QC Criteria

Platform	Library Type	Library Kit	Conc (molecules/	Conc (nM)	Conc (ng/ul)	Size(bp) From	Size(bp) To	etc
NovaSeqX	mRNA library	TruSeq stranded mRNA	-	10	-	-	-	
NovaSeqX	mRNA library	TruSeq RNA Access	-	-	5ng/ul	-	-	
NovaSeqX	mRNA library	SureSelect RNA Direct_Human	-	-	5ng/ul	-	-	
NovaSeqX	mRNA library	SureSelect RNA Direct_Mouse	-	-	5ng/ul	-	-	
NovaSeqX	mRNA library	Visium Spatial Gene Expression library	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p RNA library v3.1	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 5p RNA library v1.1 FB for ADT	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 5p RNA library v1.1 FB for HTO	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p RNA library v3.1 FB for ADT	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p RNA library v3.1 FB for HTO	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 5p RNA library v1.1	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 5p RNA library v2	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell VDJ library v2	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 5p RNA library v2 FB for ADT	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 5p RNA library v2 FB for HTO	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell Multiome Gene Expression library v1	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p Cell Multiplexing Library	-	10	-	-	-	
NovaSeqX	mRNA library	Visium Spatial Gene Expression for FFPE library	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p RNA HT Library v3.1	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p HT Cell Multiplexing Library	-	10	-	-	-	
NovaSeqX	mRNA library	SMARTer Stranded RNA library (PolyA)	-	5	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell Fixed RNA library v1 - Human 4BC	-	10	-	-	-	
NovaSeqX	mRNA library	SMARTer Ultra low input RNA library (PolyA_NexteraXT)	-	5	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p RNA HT library v3.1 FB for ADT	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell 3p RNA HT library v3.1 FB for HTO	-	10	-	-	-	
NovaSeqX	mRNA library	Chromium Next GEM Single Cell VDJ library v1.1	-	10	-	-	-	
NovaSeqX	mRNA library	BD Rhapsody Whole Transcriptome Analysis library	-	10	-	-	-	
NovaSeqX	mRNA library	BD Rhapsody Whole Transcriptome Analysis library_Enhanced Beads	-	10	-	-	-	
NovaSeqX	mRNA library	SureSelect HS2 RNA	-	-	5ng/ul	-	-	
NovaSeqX	mRNA library	Plant mRNA capture (myBaits Angiosperms-353)	-	5	-	-	-	